

# Christopher Musco

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## Education

### Massachusetts Institute of Technology

Cambridge, MA

*Ph.D. Candidate, Computer Science – Theory of Computation*

2013 – present

Advisors: Jonathan Kelner, Martin Rinard

S.M. Thesis: Dimensionality Reduction for Sparse and Structured Matrices (Spring 2015)

### Yale University

New Haven, CT

*B.S. Applied Mathematics, B.S. Computer Science*

2008 – 2012

Applied Mathematics Thesis: Fast Discrete Laplace Transforms (under Vladimir Rokhlin)

Computer Science Thesis: Graph Constructions for Machine Learning (under Daniel Spielman)

## Other Academic Positions

### Yahoo Labs

New York, NY

*Research Intern, Scalable Machine Learning Group*

Summer 2015

Mentors: Justin Thaler, Maxim Srividenko, Edo Liberty

Worked on theoretical matrix sampling problems and on algorithms for managing Yahoo's Fantasy Sports platform.

## Research Areas

theory of algorithms ◦ machine learning ◦ randomized numerical linear algebra ◦ low rank approximation  
dimensionality reduction ◦ sketching and streaming ◦ algorithmic graph theory ◦ iterative matrix algorithms

## Publications

*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science.*

Conference Papers.....

[Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition.](#)

Cameron Musco, Christopher Musco. *NIPS* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers).

[Dimensionality Reduction for K-Means Clustering and Low Rank Approximation.](#) Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *STOC* 2015.

[Principled Sampling for Anomaly Detection.](#) Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *NDSS* 2015.

[Uniform Sampling for Matrix Approximation.](#) Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS* 2015.

[Single Pass Spectral Sparsification in Dynamic Streams.](#) Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *FOCS* 2014. Invited to SICOMP Special Issue.

Other.....

[Ridge Leverage Scores for Low-Rank Approximation.](#) Michael B. Cohen, Cameron Musco, Christopher Musco. *In submission.*

[Determining Tournament Payout Structures for Daily Fantasy Sports.](#) Christopher Musco, Maxim Sviridenko, Justin Thaler. *In submission.*

## Talks

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### Randomized Block Krylov Methods

Neural Information Processing Systems (NIPS) Poster Session

Dec. 2015

### Large Scale Column Subset Selection

Yahoo Labs, Science Week Poster Session

July 2015

### Dimensionality Reduction for K-Means Clustering

IBM T.J. Watson, Research Seminar

Aug. 2015

Symposium on Theory of Computing (STOC)

June 2015

MIT, Theory Lunch

June 2014

### Principled Sampling for Anomaly Detection

MIT, Defense Advanced Research Projects Agency Site Visit

Mar. 2015

Network and Distributed System Security Symposium (NDSS)

Feb. 2015

### Uniform Sampling for Matrix Approximation

Yahoo Labs, Scalable Machine Learning Research Seminar

June 2015

### Single Pass Spectral Sparsification in Dynamic Streams

MIT, Annual Sublinear Algorithms Day Poster Session

Apr. 2015

Harvard University, Theory Seminar

Nov. 2014

Foundations of Computer Science (FOCS)

Oct. 2014

## Teaching and Mentorship

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### Massachusetts Institute of Technology

*Communication Advisor*

2015 – present

Hired to help establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively.

- Offer feedback, guidance, and coaching to individual students during weekly office hours.
- Organize outreach initiatives to increase awareness and utilization of the Communication Lab.

### Research Science Institute at MIT

*Summer Research Advisor*

2014

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program.

- Designed a project on concurrent data structures under the direction of Professor Martin Rinard.
- Chris' paper, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

### Yale University

*Teaching Assistant, Introduction to Computer Science*

2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

*Teaching Assistant, Introduction to Programming*

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 5-6 students individually.

*Undergraduate Grader, Introductory Calculus*

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

## Service

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External Reviewer.....

Symposium on Discrete Algorithms (SODA)

2016

IEEE Transactions on Signal Processing

2015

Neural Information Processing Systems (NIPS)

2015

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|-------------------------------------------------------------------------|------|
| European Symposia on Algorithms (ESA)                                   | 2015 |
| Foundations of Computer Science (FOCS)                                  | 2015 |
| International Colloquium on Automata, Languages and Programming (ICALP) | 2015 |
| Network and Distributed System Security Symposium (NDSS)                | 2015 |

## Honors and Awards

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|---------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Japan National Institute of Informatics</b><br><i>Invited participant at “Recent Advances in Randomized Numerical Linear Algebra” meeting.</i> | 2015 |
| <b>National Science Foundation</b><br><i>Graduate Research Fellowship.</i>                                                                        | 2014 |
| <b>Yale University</b><br><i>Cum Laude, Distinction in Both Majors.</i>                                                                           | 2012 |

## Professional Experience

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| <b>Redfin</b><br><i>Software Engineer, Data Team</i><br>Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.                           | <b>Seattle, WA</b><br>2012 – 2014        |
| <b>Elysium Digital</b><br><i>Summer Consultant</i><br>Provided technical expertise in software patent cases at leading litigation consulting company.                                                                                                                                            | <b>Cambridge, MA</b><br>Summer 2011      |
| <b>Amicus</b><br><i>Software Developer</i><br>Built tools for political campaign management and social fundraising at an early-stage startup that went on to raise nearly 4 million dollars in seed funding.                                                                                     | <b>New Haven, CT</b><br>2010 – 2011      |
| <b>Moody’s Investors Service</b><br><i>Intern, Asset Finance Group</i><br>Established and monitored bond ratings on securitizations in three asset classes. Wrote code to automate analysis for student loan data and to compile records on aircraft leases.                                     | <b>New York, NY</b><br>Summer 2009       |
| <b>U.S. Department of Agriculture</b><br><i>Intern, Natural Resources Conservation Service</i><br>Collected and analyzed data for two major state conservation and research projects. Integrated mapping software and database tools to develop a system for compiling and presenting soil data. | <b>East Greenwich, RI</b><br>Summer 2007 |